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# PALEOMAGNETIC RESULTS FROM NORTHERN MAINE - REINTERPRETATIONS

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Abstract. Recalculated tectonic corrections on early Devonian Dockendorff Group volcanics from northern Maine do not pass a fold test. Reflected light petrographic examination of samples from all sites reveals extensive low temperature alteration of oxides, with abundant secondary sphene and maghemite. It appears that the preserved magnetism in these rocks is post Acadian folding. The new pole position for this location is 23.5°N and 86.7°E (38 samples from 4 sites, alpha-95=7.2°). The recalculated pole position is different from published poles for both stable North American and for the proposed Avalon microcontinent.

## Introduction

An earlier report (Brown, 1979) on the paleomagnetic nature of volcanic rocks from northern Maine indicated that these rocks, assorted volcanics of the Dockendorff Group of early Devonian age, had a magnetism which pre-dated the Acadian folding. The resulting paleomagnetic pole (24° N, 132°E) was in good agreement with published data for Devonian rocks from the Avalon terrain of northern New England and the Canadian Maritime provinces. The pole did not agree with Devonian magnetic data for sediments from the stable platform of North America (see Table 2, Brown, 1979). This observation led the author to conclude that either the western boundary of "Avalon" must be further east than Presque Isle, Maine, or that northern Maine represented another sliver or slice of microcontinent, also displaced from North America in the Devonian (Brown, 1979).

## Recalculated Results

Since the publication of the earlier results, it has been noted that an error exists in the tectonic correction for site 5 (see Brown, 1979, Figure 1, Table 1). Since this site is the only one on the east side of the large syncline in which the units outcrop, its corrected position is crucial to the final results. The new data, after tectonic corrections for a vertical unit in a southward plunging syncline, are presented in Table 1. The recalculated means for all sites and samples are presented in Table 2. With the corrected data for site 5, it is evident that this population does not pass the fold test, and the paleomagnetic data is more consistent before unfolding.

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## Oxide Mineralogy

Reflected light petrographic examination of samples taken from all sites investigated in this study reveals a large amount of post cooling, low temperature alteration. The primary Fe-Ti oxide phases in these samples were magnetite and ilmenite ranging in size from 0.5 - 1.5 mm. The crystals are subhedral to anhedral. A few samples exhibited textures indicative of small amounts of high temperature (above 600°C) subsolidus oxidation exsolution.

Secondary alteration has produced maghemite in place of much (30-50%) of the magnetite. In some samples all of the magnetite has gone to maghemite. Additionally, alteration of Fe-Ti oxides to sphene is extensive in these rocks. Sphene occurs as pseudomorphs after ilmenite and magnetite in approximately 50% of the samples examined. In a few cases, all of the oxides were completely sphenitized. Minor amounts of secondary, i.e. post cooling, hematite occur in a few samples. Small amounts of primary pyrrhotite were observed in one specimen.

## Discussion

From the site and sample mean data presented in Table 2, it is evident that these rocks are carrying a magnetism indicative of post-folding. Folding in this area is Acadian in age (Boucot, et al., 1964; Pavlides, 1974) so that the pole position must represent a post-Acadian (post mid Devonian) position.

The magnetic mineralogy supports this with the indication that extensive alteration has occurred and that the major remanence is secondary in nature. The formation of sphene from existing iron oxides is commonly related to low grade regional metamorphism (Haggerty, 1976), but it affects the natural remanence by decreasing the intensity only. The oxidation of magnetite to maghemite, well documented in these rocks, is another low temperature alteration and could have occurred during or after the folding phase of the Acadian orogeny. The effect of this alteration on the natural remanence to produce a CRM, which direction can be variable with respect to the original direction (Merrill, 1975), but in many instances can be very stable. Therefore, it seems likely that the magnetization measured in the Dockendorff Group was formed (or altered) in the Acadian orogeny and represents a middle Devonian pole position.

Comparison of this pole to average poles for middle and upper Devonian rocks of stable North American (45°N, 122°E) and Upper Devonian rocks of

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TABLE 1. MAGNETIC DIRECTIONS BEFORE AND AFTER TECTONIC CORRECTIONS

| | N/S | H | I | D | I' | D' | k | α-95 |
|---|---|---|---|---|---|---|---|---|
| Site 2 (N10E, 90E) | 8/9 | 20 | 27.8 | 232.2 | 32.4 | 151.7 | 11 | 17.1 |
| Site 4a (N10E, 50E) | 14/14 | 20 | 42.8 | 202.5 | 33.8 | 159.1 | 96 | 4.1 |
| Site 4b (N10E, 50E) | 5/5 | 20 | 34.7 | 204.1 | 31.1 | 168.5 | 102 | 7.6 |
| Site 5 (NS, 20S) (N155E, 90W) | 11/16 | 20 | 11.6 | 187.1 | -29.0 | 145.8 | 20 | 10.6 |

S is the number of samples collected, N is the number of samples used in site means
H is the demagnetization level in milliteslas
I and D are the magnetic inclination and declination before tectonic corrections
I' and D' are the magnetic inclination and declination after tectonic corrections
k and α-95 are the precision parameter and radius of cone of 95% confidence about the mean
Tectonic corrections (direction and angle of rotation) are given in parenthesis for each site

the Avalon Platform (28°N, 120°E) are given in Table 3. The Dockendorff pole has a low latitude, as does the Avalon pole, but it is displaced considerably to the west. A similar pole nas been determined for the early Devonian Traveler felsite (Spariosu and Kent, 1980; Table 3), located 100 km southwest of the Dockendorff outcrops.

Interpretation of this new data leads us to suggest that northern Maine was separate from stable North American in the middle to upper Devonian. However, its relation to Avalon is not clear. It is possible that this area was separate from Avalon, or was part of the Avalon. Platform and underwent subsequent rotation. The

TABLE 2. MEAN MAGNETIC DIRECTIONS BEFORE AND AFTER TECTONIC CORRECTIONS

| | N | I | D | k | α-95 | Pole | Position |
|---|---|---|---|---|---|---|---|
| Folded | | | | | | | |
| Mean Sites | 4 | 28.7 | 206.0 | 15 | 24.9 | 24°N | 84°E |
| Mean Samples | 38 | 28.7 | 203.6 | 11 | 7.2 | 24°N | 87°E |
| Unfolded | | | | | | | |
| Mean Sites | 4 | 16.3 | 156.1 | 7 | 38.4 | 13°N | 140°E |
| Mean Samples | 38 | 14.2 | 155.1 | 6 | 10.3 | 32°N | 141°E |

N is the number of samples or sites
I and D are the magnetic inclination and declination
k and α-95 are the precision parameter and radius of cone of 95% confidence about the mean

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TABLE 3. MID-PALEOZOIC PALEOMAGNETIC POLES, EASTERN NORTH AMERICA

| Location | Age | Pole | Reference |
|---|---|---|---|
| Eastern North American Platform (N = 6) | m-uD | 45°N, 122°E | Brown, 1979 |
| Avalon Platform (N = 5) | uD | 28°N, 120°E | Brown, 1979 |
| Traveler felsite, Me | 1D | 29°N, 82°E | Sparlosu and Kent, 1980 |
| Dockendorff GP, Me | mD | 24°N, 84°E | this study |

suggestion that northern Maine was a separate sliver of microcontinent is not confined to paleomagnetic data. Osberg (1978) made such a suggestion on the basis of basement rocks, although in his interpretation affinities with North America was reached prior to the Devonian. It is obvious from this study, and comparison to other paleomagnetic work, that much more information is needed before interpretation of the position of northern Maine in the Paleozoic is well understood.

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